DEVELOPMENT CONFERENCE PACKAGE

Supplementary Appendix

Development methods, evidence, and reproducibility map

NOT FINAL EVIDENCE | NOT AUTHORIZED FOR SUBMISSION

Evidence cutoff: 29 August 2026

S1. Endpoint boundaries

Environment	Target	Primary-paper role	Prohibited interpretation
PIDD	Configured diabetes-positive label	Primary, separate	Universal-population risk or future onset
Sylhet	Early-stage diabetes-positive label	Primary, separate	Incidence prediction or transport beyond symptom cohort
BRFSS 2015	Self-reported diagnosed-diabetes status	Primary, separate	Clinical diagnosis or future incidence
Hospital130	Readmission within 30 days	Secondary, separate endpoint	Diabetes-status prediction
Mixed router	No shared target	Software demonstration	Cross-cohort probability fusion

S2. Development protocol

The development profile uses seed 42, 30 hyperparameter trials, 500 bootstrap resamples, average precision as the tuning objective, and identity, sigmoid, and isotonic calibration candidates. Isotonic calibration requires at least 1,000 calibration observations. Threshold policies are fixed 0.5, development Youden, and a sensitivity-constrained threshold targeting 0.80 sensitivity.

PIDD and Sylhet use repeated nested validation. PIDD uses five repeats, five outer folds, and five inner folds. Sylhet additionally groups identical feature profiles. Large datasets use concealed manifest-based holdouts. Final profile execution is blocked in software until authorization metadata and a frozen protocol are present.

S3. Metric taxonomy

Domain	Metrics	Interpretation rule
Probability accuracy	Brier, scaled Brier, log loss	Primary reliability evidence; lower is better
Discrimination	AUROC, average precision	AP is prevalence-sensitive and is always paired with prevalence
Calibration	Intercept, slope, O:E, ECE, curves	A percentage is not treated as reliable from AUROC alone
Classification	Sensitivity, specificity, PPV, NPV, F1, MCC, balanced accuracy	Secondary and threshold-dependent
Diagnostic	LR+, LR-, diagnostic odds ratio	Descriptive; undefined values handled explicitly
Clinical utility	Net benefit over threshold range	Compared with treat-all and treat-none
Applicability	Supported-panel coverage, abstention rate and reason	Reported with every incomplete-input analysis

Domain	Metrics	Interpretation rule
Selective prediction	Risk-coverage, event coverage, accepted prevalence	Gains are invalid if achieved by hidden disproportionate rejection

S4. PIDD artifact-completeness audit

Experiment pidd_development_072cd2 104 089 contains 25 outer split pairs for each of five model families, 19,200 held-out fold predictions, 3,840 aggregated participant-model rows, complete tuning records and saved models, 20 primary summary rows, and no non-finite numeric values in the primary summary. Metadata records the complete 25-split manifest, five inner folds, model list, raw-data hash, dataset-configuration hash, source-tree hash, and participant-level inference scope.

Dependent evidence directories include availability routing, matched panel contrasts, artifact ablations, missing-indicator retraining, selective prediction, subgroup analysis, missingness stress, explanations, and reporting ablations. Panel-specific retraining is complete across all 25 outer splits for the prespecified Random Forest and XGBoost reduced-panel comparisons.

S5. Applicability and abstention contract

Prediction proceeds in this order:

1. Parse dataset-qualified feature names and reject unknown or conflicting keys.
2. Validate type, coding, plausible range, and required endpoint contract.
3. Identify supported panels within each dataset router.
4. If no route is eligible, abstain with a machine-readable reason.
5. If multiple incompatible environments remain eligible, require explicit dataset selection or abstain; do not fuse probabilities.
6. Load the saved bundle for the selected panel and fold ensemble.
7. Apply fitted preprocessing and calibration.
8. Return the label probability, uncertainty/provenance fields, selected dataset, panel, model family, calibration method, and threshold interpretation.

S6. Ablation inventory

Ablation	Scientific question	Evaluation control
Calibration selected vs raw	Does calibration improve probability accuracy?	Same outer predictions and participants
Router vs robust fixed model	Does model switching add value?	Same panels and participants
Brier vs AP/log-loss routing	Is routing objective consequential?	Same candidate model bank
Tree models vs Logistic Regression	Is complexity justified?	Paired held-out participants

Ablation	Scientific question	Evaluation control
Missing indicators on vs off	Do indicator features add value?	Separate tuning; same splits, models, budgets, seeds
Panel-specific fit vs masked full model	Does training for a panel outperform masking?	Same split, panel, family, budget, participant
Named panels vs size-only heuristic	Is panel composition more informative than count?	Regression on held-out panel results
Grouped vs encoded SHAP	Does clinical grouping change interpretation?	Same held-out SHAP values
Leakage demonstrations	How optimistic are invalid designs?	Explicitly prohibited from efficacy selection

S7. PIDD development robustness summary

The five full-panel models occupied a narrow performance range. The full-panel availability router and robust fixed baseline were effectively tied at the reported precision. No matched reduced-panel comparison survived Holm correction. The core-metabolic panel retained most probability performance, while the non-laboratory and glucose-plus-age panels showed larger descriptive loss.

The full missing-indicator retraining study found no corrected benefit for indicator-on Logistic Regression or Random Forest across AUROC, average precision, Brier score, or log loss. Adjusted p-values ranged from 0.200 to 1.000, and all paired intervals crossed no improvement.

Panel-specific Random Forest and XGBoost fitting was evaluated for five reduced panels over all 25 outer folds, producing 76,800 prediction rows and 40 paired metric contrasts. Several unadjusted intervals favoured panel-specific fitting, but no comparison survived Holm correction; the minimum adjusted p-value was 0.120. The evidence does not justify a claim that retraining a specialist model is generally superior to masked full-panel inference.

Selective probability-certainty acceptance improved retained-case Brier score as coverage decreased, but retained positive events less often than negative cases. Age-group coverage was heterogeneous and the oldest group was too small for stable performance estimation. Severe simulated missingness raised probability loss and left fewer than 7% of participants compatible with a named panel. These results favour explicit abstention over extrapolation.

S8. Reporting and risk-of-bias controls

The full grouped Sylhet benchmark (sylhet_development_b52 477 910fe6) contains 25 grouped outer splits per model, complete aggregation for all 520 participants, cluster-bootstrap intervals, 20 primary summary rows, and no non-finite primary metrics. The full router achieved participant-level AUROC 0.965 and Brier 0.068, but no routing, model-family, calibration, selector-metric, or matched-panel comparison survived Holm correction. Extreme calibration slopes, suppressed subgroups, and severe-dropout applicability collapse are retained as limitations rather than hidden behind the cohort's high discrimination.

Full Sylhet panel-specific retraining covered four reduced panels, two model families, and all 25 grouped outer splits. It produced 41,600 held-out prediction rows and 32 paired metric contrasts. No contrast survived Holm correction; the minimum adjusted p-value was 0.064. The full missing-indicator retraining family also produced no corrected benefit: Random Forest was numerically equivalent, and all Logistic Regression adjusted p-values were 1.000. Neither component supports a model-superiority claim.

The final package must include a completed TRIPOD+AI checklist and a reviewed PROBAST+AI assessment. Key safeguards already implemented are fold-local preprocessing, nested selection, participant-level repeated-CV aggregation, profile-cluster resampling for Sylhet, patient grouping for Hospital130, outcome-blind manifests, calibration evaluated on held-out data, paired comparisons, multiplicity correction, complete artifact provenance, and fail-closed final authorization.

Residual concerns are public-benchmark reuse, small cohorts, self-reported BRFSS status, limited transportability, missingness-mechanism uncertainty, subgroup sample size, and absence of external or temporal validation. These limitations must remain visible even if final internal metrics are strong.

S9. BRFSS full-development evidence

Study brfss_development_holdout_56 407ef53bba used the full 429,360-row retained cohort with the existing seed-42 development contract: 343,488 rows across five development folds and 85,872 rows in the development test role. The latter contained 11,451 recorded positives. Because seed-42 outcomes were inspected in prior development work, this evidence is not eligible for final claims.

All five model families completed 30 development tuning trials, cross-fitted calibration selection, three threshold policies, 500 participant-bootstrap intervals, paired comparisons, paired decision-curve inference, bundle export, and exact reload verification. Development OOF selection chose LightGBM. On the development test, XGBoost had calibrated AUROC 0.8430, average precision 0.4436, Brier 0.09150, and log loss 0.29581; CatBoost and LightGBM differed by less than 0.0002 AUROC and 0.00005 Brier. No top-three pairwise metric difference survived Holm correction.

The availability router used LightGBM for the 21-feature panel and XGBoost for all five reduced panels. Full-panel Brier was 0.09154; the two-feature panel had Brier 0.10198. None of 80 matched full-versus-reduced contrasts survived metric-wise Holm correction (minimum adjusted p=0.0798), although every pointwise degradation interval excluded zero. Matched-panel contrasts, artifact ablations, certainty-based selective prediction, subgroup analysis, survey sensitivity, and held-out SHAP were executed using the supported holdout paths. Retraining-only missing- indicator, panel-training, simulated-missingness, and reporting-ablation engines currently apply to repeated nested-CV studies and were not misrepresented as BRFSS holdout analyses.

Survey weighting yielded 10.95% prevalence, compared with 13.34% unweighted. For XGBoost, weighted AUROC was 0.8707 and Brier 0.07517. The 500-replicate stratified PSU bootstrap produced pointwise intervals, but the random respondent holdout was not designed as an official survey partition; this remains a secondary sensitivity analysis.

At 70% full-panel selective coverage, Brier score improved to 0.0430, but positive-event coverage collapsed to 25.0%, accepted prevalence to 4.76%, AUROC to 0.784, and average precision to 0.128.

These values demonstrate selective case-mix change rather than clinical utility and must be presented together.

The BRFSS subgroup analysis estimated 30 age-group and sex cells across five coverage targets, with no small-cell suppressions under the configured minimums of 50 accepted observations and 10 positive and 10 negative events. At the 70% target, observed subgroup coverage ranged from 69.7% to 70.3%; accepted prevalence ranged from 4.59% to 4.75% and average precision from 0.125 to 0.133. These are within-cohort descriptive estimates, not evidence of population representativeness or causal fairness.

In the BRFSS model-family ablation, each tree model improved over Logistic Regression on AUROC, average precision, Brier score, and log loss after metric-family Holm correction (adjusted p=0.016). Calibration improved CatBoost Brier score by 0.07762 and log loss by 0.19890 relative to its uncalibrated raw score; LightGBM selected identity calibration and showed no change. Router selection by average precision or log loss did not yield corrected improvements over the prespecified Brier selector, and routing did not outperform the robust single-model comparator after correction. These are development ablations.

Held-out SHAP explanations used 1,000 test-role respondents per model and a 100-row development-only background. Original-feature aggregation placed general health, high blood pressure, age, BMI, and high cholesterol in the top five for four models; CatBoost contained the same features in a different order. Maximum additive reconstruction error ranged from 2.2e-15 for Logistic Regression to 5.6e-6 for XGBoost. The holdout design supplies one explanation sample rather than fold-pair stability estimates, and SHAP values are not causal effects.

S10. Reproducibility artifact map

Artifact	Location
Development configuration	config/experiments/development.yaml
Publication controls	config/publication_protocol.yaml
Statistical analysis plan	docs/research/Statistical_Analysis_Plan.md
Endpoint hierarchy	docs/research/Endpoint_and_Claim_Hierarchy.md
Leakage/applicability audit	docs/research/Data_Leakage_and_Applicability_Audit.md
Decision log	docs/research/Decision_Log.md
PIDD development study	outputs/publication_pipeline/studies/pidd_development_072cd2104089
Sylhet development study	outputs/publication_pipeline/studies/ sylhet_development_b52477910fe6
BRFSS development study	outputs/publication_pipeline/studies/ brfss_development_holdout_56407ef53bba
Tracked study registry	evidence/study_registry.json
Data-source manifest	config/data_sources.yaml
Exact environment lock	requirements.lock
Command reference	docs/RUNBOOK.md

Artifact	Location
Automated tests	tests/

S11. Items reserved for locked evidence

- Final participant-flow counts, exclusions, and the protected authorized BRFSS decision table from a new concealed partition, not the seed-42 development test.
- Frozen model/panel choices and justified non-inferiority margins.
- Authorized concealed estimates with 2,000-resample intervals, decision curves, calibration plots, subgroup tables, and the claim-evidence map.
- Repository release tag, archive DOI, author list, affiliations, funding, conflicts, ethics wording, and venue-specific formatting.